Router Interrogation Scene Guide

Windows + Kali WSL Workflow

Delaware County High Tech Crime Unit

Purpose

This guide is designed for on-scene router interrogations during:

- Search warrants
- Consent searches
- Missing person investigations
- Homicide investigations
- ICAC investigations
- General criminal investigations

The router is rarely the evidence itself.

The router is often a witness.

The goal is to determine:

- Who was present
- What devices were present
- When devices arrived
- When devices left
- What other evidence exists
- What evidence should be seized next

Before Entering the Network

Before connecting anything:

Observe.

Document.

Photograph.

Do not immediately start typing commands.

Look at the environment first.

Initial Scene Assessment

Identify:

- Router
- Modem
- Mesh nodes
- Network switches
- Wireless access points
- DVRs
- NVRs
- NAS devices
- Smart home hubs
- Attached USB storage

Take photographs of:

- Front of router
- Rear of router
- Model labels
- Serial numbers
- Connected cables
- Nearby network equipment

Document:

- Date
- Time
- Investigator
- Case number
- Router manufacturer
- Router model
- Router serial number
- ISP if known

Determine Volatility

Ask yourself:

What information will disappear if this device loses power?

Many routers only retain:

- DHCP history
- Wireless history
- DNS logs
- System logs
- Administrative logs

for a limited period of time.

If the router is operating normally, avoid unnecessary reboots.

Windows Collection

Connect your examination laptop.

Run:

```
python router_windows_collect.py
```

This creates the case folder and documents:

- Examiner system information
- Network configuration
- Routing information
- ARP cache
- Active connections
- DNS cache
- Wireless configuration
- Adapter information
- Router connectivity

Review:

WINDOWS_COLLECTION_SUMMARY.txt

Identify:

- Default gateway
- DNS servers
- Local subnet
- Connected network adapters

Kali WSL Collection

Launch Kali WSL.

Run:

sudo ./router_ir_collect_v2.sh <case_folder>

Example:

sudo ./router_ir_collect_v2.sh /mnt/c/Users/Kris/Desktop/Router_IR_20260531_120000

The script automatically collects:

- Network configuration
- Routing information
- Neighbor table
- Router banners
- Router SSL information
- ARP scan
- Netdiscover results
- Nmap discovery
- Router service enumeration
- Packet capture
- SHA256 hashes

Review:

ROUTER_SUMMARY.txt

This becomes your initial network intelligence report.

Identify The Router

Review:

default_gateway.txt

Determine:

- Router IP
- Router vendor
- Router services
- Router operating system clues

Review:

nmap_router_services.nmap

Look for:

- HTTP
- HTTPS
- SSH
- Telnet
- SNMP
- UPnP
- VPN services

These services may reveal additional access methods or administrative interfaces.

Identify Devices Present

Review:

- arp_scan.txt
- netdiscover.txt

nmap_host_discovery.nmap

You are attempting to identify people through devices.

Examples:

- Kris-iPhone
- Samsung-S24
- Titus-iPad
- RingCam01
- HP-Laptop

Ask:

Who owns this device?

Create a list of:

- Known devices
- Unknown devices

Unknown devices often become investigative leads.

Identify Additional Evidence Sources

The router often points investigators toward more important evidence.

Look for:

- Cameras
- DVRs
- NVRs
- NAS devices
- Smart home systems
- Cloud-connected devices

The router frequently serves as a roadmap to additional evidence.

Router Login

Using a browser:

- http://ROUTER-IP

https://ROUTER-IP

Immediately document:

- Router time
- Router timezone
- Router uptime
- WAN IP
- Firmware version

This information becomes critical during timeline reconstruction.

High Priority Evidence Collection

If time is limited, collect these first.

DHCP History

Export:

- Current leases
- Historical leases

Questions:

- Was the target device present?
- When was it assigned an address?
- When did it disappear?

Wireless History

Export:

- Current wireless clients
- Historical wireless clients

Questions:

- When did the target device connect?
- When did it disconnect?
- When was it last seen?

Connected Devices

Export:

- Device list
- Offline devices
- Known devices
- Guest devices

Questions:

- Who was present?
- Who may have been present?

Router Logs

Export:

- System logs
- DHCP logs
- Wireless logs
- DNS logs
- Firewall logs
- Administrative logs
- VPN logs

These logs frequently contain the timeline.

Administrative Activity Review

Search exported logs for:

- login
- logout
- admin
- authentication
- failed

- password
- configuration
- firmware

reboot

Questions:

- Was the router accessed?
- Was the router modified?
- Was the router rebooted?

Administrative activity near the incident may be significant.

Remote Access Review

Review:

- Port forwarding
- UPnP
- Remote management
- SSH
- Telnet

Questions:

- Could someone access this network remotely?
- Could devices inside the residence be reached from the Internet?

VPN Review

Search for:

- wireguard
- openvpn
- ipsec
- tailscale

zerotier

Questions:

- Was a VPN configured?
- Was a VPN active?

Remember:

VPN presence is not suspicious by itself.

It is simply part of understanding the network.

Cloud and Location Services Review

Search logs for:

- icloud
- findmy
- google
- maps
- gmail
- dropbox
- onedrive
- discord
- telegram
- signal
- snapchat
- facebook
- instagram

tiktok

Questions:

- What cloud services were used?
- What communication platforms were used?
- Are there additional warrant opportunities?

Packet Capture

If live traffic is important:

Use Wireshark on Windows.

Capture:

```
host ROUTER-IP
```

or

```
net 192.168.1.0/24
```

Collect:

- ARP
- DHCP
- DNS
- Router traffic
- Device communications

Save capture into the case folder.

Hash after collection.

Build the Presence Timeline

Attempt to answer:

- What devices were present?
- When were they present?
- When did they leave?
- Did new devices arrive?
- Did the router reboot?
- Was the router administered?
- Were cameras operating?
- Were VPNs active?

This is where the router becomes valuable.

Missing Person Investigations

Primary questions:

- Was the phone connected?
- When was it last connected?
- Did it leave the network?
- Did another device arrive afterward?
- Were mapping or location services used?

The goal is often determining the last known presence of a device.

Homicide Investigations

Primary questions:

- Who was present?
- What devices were present?
- Were cameras operating?
- Were devices removed afterward?
- Did the router reboot?

The goal is often establishing presence and movement.

Child Exploitation Investigations

Primary questions:

- What devices are associated with the residence?
- What users appear associated with those devices?
- Were cloud services used?
- Were VPNs configured?
- Are additional devices present that require seizure?

The goal is identifying all potential evidence sources.

Before Leaving The Scene

Do not leave until you can reasonably answer:

- What router is present?
- What devices are present?
- What devices were present historically?
- Was the target device present?
- Were cameras identified?
- Were logs exported?
- Was the configuration preserved?
- Was router time documented?
- What evidence should be seized next?